# Multi-Stage Transfer Learning for Efficient Video Prediction and Segmentation with Sparse Labeled Data

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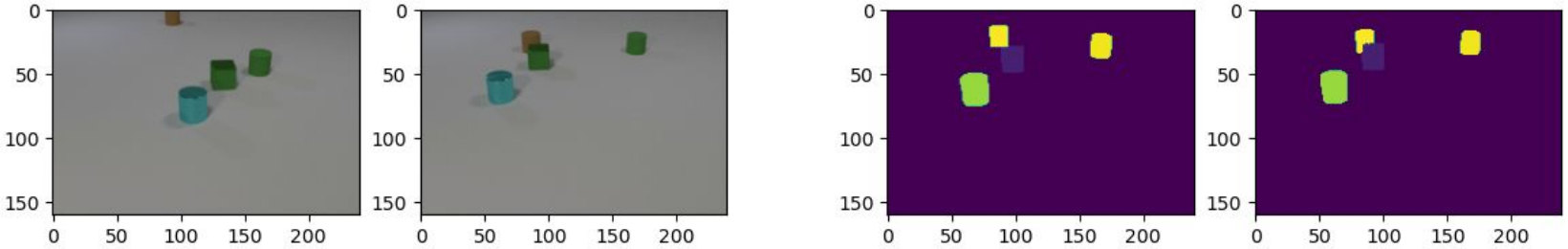
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# Introduction

Task:

- Video prediction predict frame[11:22]
- Segmentation: predict mask[22]



# Challenge: scarce labeled data for segmentation

Dataset

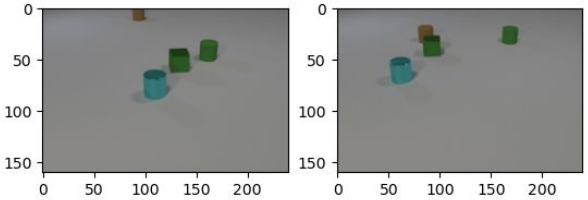
- Labeled training set: **1000** videos (11+11 frames and corresponding masks)
- Unlabeled training set: **13000** videos (11+11 frames and no masks)
- Validation set: 1000 videos (11+11 frames and corresponding masks)
- Test set  (hidden)

Problem: expensive to label each frame!

# Remedy: Multi-Stage Transfer Learning

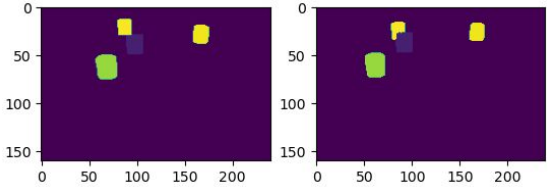
## Stage 1:

13000 unlabeled




UNET

13000 masks




(Trained with 1000 labeled )

## Stage 2:

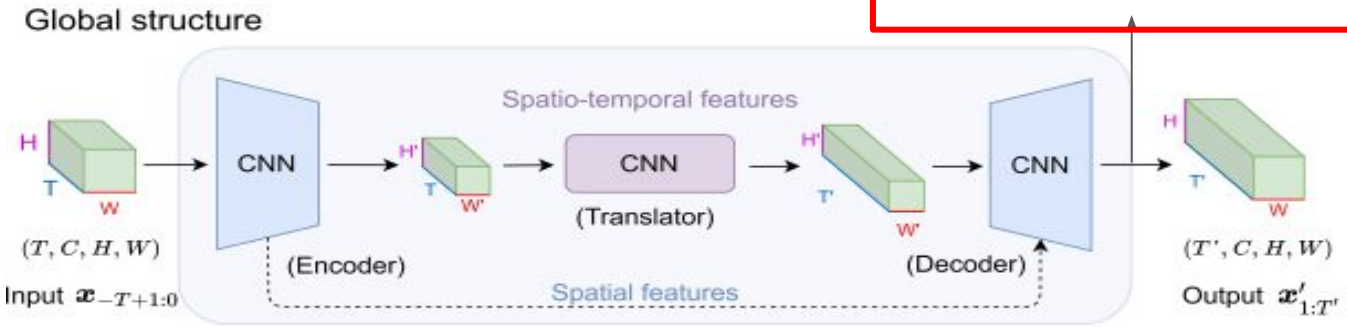
Input frame[:11] → Modified SimVP → Prediction of frame[11:22] + Segmentation of frame [22]

(Trained with combined dataset)

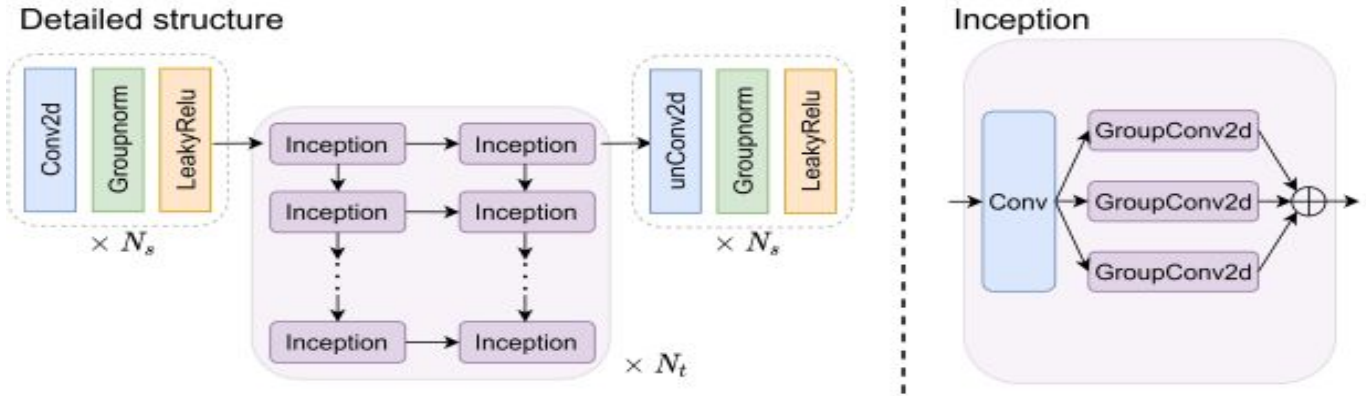
# Detail for Stage 2: SimVP for prediction

**Epoch 31 - 60**

**(T' , num_classes, H, W)**




**Epoch 0 - 30**




Zhangyang Gao, Cheng Tan, Lirong Wu, Stan Z. Li. "SimVP: Simpler yet Better Video Prediction". CVPR, 2022.

# Our Contribution

- Enrich the dataset by a multi-stage transfer learning strategy.

- Modify the SOTA video prediction algorithm (SimVP) to achieve end-to-end prediction + segmentation.